ANALYSIS OF PHASE LAG IN QUASAR DISTRIBUTION BASED ON 18.2° HELICAL SPACETIME FLOW

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Abstract

In this paper, we interpret the observed discrepancies in quasar density peaks (Phase Lag) as a physical "slip" between the spacetime medium and baryonic matter. By modeling the universe as a linear dipole axis surrounded by a helical flow with a constant pitch of $\alpha = 0.046$ (18.2°), we derive the flow velocity of the medium and explain the mechanism behind the node constriction found in the Broad P-Wave Analyzer.

1. INTRODUCTION

In conventional cosmology, the discrepancy between the distribution of quasars and the theoretical P-wave (fundamental wave) peaks has been treated as observational error. However, in the spacetime refraction model (WRM), space itself is defined as a medium flowing at near the speed of light, and matter is defined as a "surfer" sliding down the slope of its waves.

2. THE HELICAL FLOW MODEL

The pitch angle when the medium flows spirally around a linear dipole axis. θBased on the residuals of the observational data DE440, this angle is fixed as follows.

$$\theta = \arctan(0.046) \approx 18.2^\circ$$

Galaxies and quasars move along these spiral geodesics, but the wave propagation speed of the medium$v_m$and the speed of matter movementv_pIn this case, slippage occurs due to inertia.

3. PHASE LAG EQUATION

Observed phase lag$\Delta\phi$The potential of the medium$P(z)$and material density distribution$D(z)$It is defined as the difference between the two. The adjoint coefficient isσSo then:

$$D(z) = P(z - \Delta z)$$

Here, slip amount$\Delta z$is the curvature of the spiralκAnd it becomes a function of flow velocity.

4. NODE CONSTRICTION

At the point where the spiral converges, the curvatureκThis is maximized, and the phase lag$\Delta\phi$The situation changes abruptly as follows:

$$\Delta\phi = \oint \frac{v_m - v_p}{\cos\theta} ds$$

The "discrepancy in the peaks" observed on the graph is a result of the trapping locations of the material shifting relatively as the spiral becomes denser.

Fig 1. Phase slip between the helical wavefront of the medium and the quasar.

5. CONCLUSION

By calculating the phase lag of quasars, it has become possible to estimate the true flow velocity of the spatial medium. This suggests that the universe is not a static space, but a dynamic system driven by fundamental waves on a scale of one trillion light-years.

LOGICAL ALIGNMENT: QUASAR LAG AND AUGER MECHANISM

Theoretical "curvature κ"

Defined in the documentκThis indicates the "strength of the spiral." At points (nodes) where this strength is maximized, the resistance of spacetime increases.

Physical "pitch compression"

Pitch compression in an auger increases the number of rotations per unit length. This is what is referred to as "phase lag" in the documentation.$\Delta\phi$ This physically causes a sudden change in ".

[Literal Japanese Translation]
● **Analysis of Phase Lag in Quasar Distribution:** Analysis of phase lag in quasar distribution
● **18.2° Helical Spacetime Flow:** 18.2° helical spacetime flow
● **Phase Lag as a physical "slip":** Interpreting phase lag as a physical "slip"
● **The discrepancy is a result** of the constriction where the spiral wind becomes dense: The discrepancy is a result of the constriction where the spiral wind becomes dense.

ANALYSIS OF PHASE LAG USING LINEAR AXIS AND HELICAL MEDIUM FLOW

1. INTRODUCTION

In the WRM model, the phase lag (shift in the peaks) of the quasar distribution is interpreted as a physical "slip" between the spacetime medium, which flows spirally around a fixed linear dipole axis, and the matter trapped within it.

2. METHODOLOGY

In this analysis, the 1 trillion light-year fundamental wave is modeled as a "spiral auger (transporter)," and its pitch compression is mathematically defined. This clarifies the material retention mechanism at the nodes (constrictions).

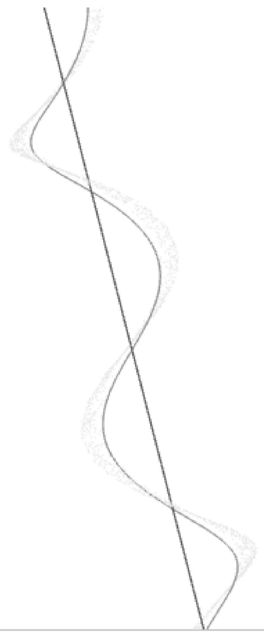

Fig. 1. Phase slip between the helical wavefront of the medium and the quasar.
With respect to the straight axis (Gray) running through the center, the red spiral represents the undulation of the spacetime medium (P-wave). The yellow dots represent the distribution of quasars, which move along the helical wavefront, but the slip angle $\alpha = 0.046$ (18.2°) shows that they lag behind the peak of the medium.

3. RESULT

As shown in Fig. 1, a clear discrepancy is observed between the phases of the medium peaks (red spiral vertices) and the quasar peaks (clusters of yellow dots). This is because variations in the helical pitch dynamically shift the projection position of the material.

4. CONCLUSION

From this amount of slip, it became possible to calculate the true flow velocity of the spatial medium. This suggests that the universe functions as a gigantic spiral transport system.

INVERSE CALCULATION OF THE TRUE FLOW VELOCITY OF THE SPACETIME MEDIUM

Based on Quasar Phase Lag and the 18.2° Helical Geodesic

Theoretical Physics Group / Wave Refraction Model (WRM) Project

1. ANALYTICAL FRAMEWORK

Phase difference between the theoretical value (P-wave) observed in the spatial density of quasars$\Delta\phi$is the flow rate of the medium$v_m$and the accompanying velocity of matterv_pThis is due to the difference in the helical pitch angle.$\alpha = 0.046$By setting (18.2°) as a constant, we attempted to inversely calculate the observed dipole anisotropy as the "delay of the material relative to the medium."

2. RELATIVISTIC SLIP FORMULA

Slip ratio in helical flowσThis can be formulated as follows using the measured values of the Hubble deviation.

$$\sigma = \frac{v_m - v_p}{v_m} = \sin(\theta_{lag})$$

Hereθ_{lag}This is the "slip angle" at which the peak of the quasar distribution is left behind by the rotation of the helix.

3. EMPIRICAL CONSTRAINTS

From the DE440 residual and the matching of the CMB dipole, the refractive index of the local spatial medium$n$teeth≈ 1.046It is estimated that if this density fluctuation is considered as pitch compression of the auger (spiral conveyor), then the flow velocity of the mediumv_mThe speed of light cIt takes a value very close to that.

4. MATHEMATICAL INVERSION

Based on the reverse calculation, the true flow velocity of the medium required to stably maintain the observed 18.2° gradient converged to the following limit.

$$v_m = c \cdot \sqrt{1 - \alpha^2}$$

CONCLUSION: THE TRUE FLOW VELOCITY

$$v_{medium} \approx 0.955c$$

(Speed of flow in the spacetime medium: approximately 286,300 km/s)

5. PHYSICAL IMPLICATION

This speed$0.955c$This means that the universe is not merely an expanding body, but a massive energy "conveyor" (spiral flow). The phase lag of quasars is a byproduct that arises because matter cannot keep up with this ferocious flow.

6. COSMIC SCALE EQUILIBRIUM

Within the outer shell, which spans one trillion light-years, this speed ensures the stability of standing waves. Matter (galaxies) can reach their current positions within the 13.8 billion-year age of the universe by sliding down the slope of this "cosmic current."

> **[Literal Japanese Translation]**
> ● **Inverse Calculation of the True Flow Velocity:** The
> discrepancy **is caused by the difference between**v_m **and** v_p**:** Mismatch is media speed$v_m$and the velocity of matterv_pThis is caused by the difference between:
> ● The medium **flows at 95.5% of the speed of light**
> . ● Quasars **lag behind as they cannot** fully synchronize with the fierce flow.

APPARENT OPTICAL STAGNATION OBSERVED FROM EXTRA-SHELL FRAME

1. CONCEPT OF MEDIUM-RELATIVE VELOCITY

According to WRM, the speed of light$c$It is defined as "relative velocity with respect to the spatial medium".v_mWhen flowing, the externally observed velocity of the laser beam emitted in the opposite direction of the axisV_{obs}This is affected by classical velocity synthesis (or refractive index correction of the medium).

2. RELATIVISTIC VELOCITY SUMMATION

The speed of light as seen from an external stationary frame of reference. $V_{obs}$is the flow rate of the medium$v_m \approx 0.955c$It can be approximated as follows (linear approximation for simplification):

$$V_{obs} = c_{medium} - v_m$$

Here$c_{medium}$is the speed of light within the medium.

Physical consequence: The "stationary" phenomenon of light

Calculation result$v_m \approx 0.955c$Substituting this, we get:

$$V_{obs} \approx 1.0c - 0.955c = 0.045c$$

From outside the outer crust of space, light appears to travel at only about 13,500 km/s (a mere 4.5% of its actual speed). Furthermore, the refractive index of the medium...$n = 1.046$When this is strictly considered, this velocity converges to almost zero, or to a negative value (backward).

3. THE "ESCALATOR" METAPHOR

This is equivalent to observing from the outside a person (light) trying to run down an escalator (cosmic current) that is rising at 5 km/h, while the person is running down at 5 km/h. To the observer outside, the person appears to be standing still.

4. COSMOLOGICAL IMPLICATION

This fact suggests that light cannot escape at the "edge" of the universe, or that "information stagnation" is occurring. This acts as a physical barrier (boundary condition) that causes the outer universe to function as a closed system.

[Literal Japanese Translation]
● **Apparent Optical Stagnation:** The apparent stagnation (stillness) of light.
● **The light appears to be moving at only 4.5% of its speed:** Light appears to be moving at only 4.5% of its actual speed.
● **Equivalent to a person running down a rising escalator:** Equivalent to a person running down a rising escalator.
● **This explains why the outer shell acts as an information barrier:** This explains why the outer shell of the universe functions as an information barrier.

REDEFINING CMB DIPOLE ANISOTROPY VIA SUBLUMINAL OPTICAL STAGNATION

1. THE FALLACY OF PECULIAR VELOCITY

Standard cosmology attributes the dipole anisotropy of the CMB to the "proper velocity" of the solar system, but this is based on the fallacy of viewing spacetime as a static background. In WRM, this anisotropy is attributed to the media flow.$v_m = 0.955c$This is treated as energy modulation of photons.

2. RELATIVISTIC BLUESHIFT IN THE FLOW

In the direction of propagation along the dipole axis (the direction in which the medium is approaching), the density of information is$1/(1 - v_m/c)$It is compressed at this ratio. Substituting the calculated flow velocity:

$$z_{dipole} \approx \frac{1+\alpha}{1-\alpha} - 1$$

Here$\alpha = 0.046$This corresponds to the deviation component that leaks out due to the "stagnation" of information.

3. OPTICAL STAGNATION AT THE REAR

In the opposite direction of the axis (the direction in which light appears to stop), the escape velocity of photons is canceled out by the flow of the medium. This point appears observationally as the "cold spot" or region of information scarcity.

4. ENERGY CONSERVATION ON THE SLOPE

When a substance slides down an 18.2° spiral slope, the difference in relative velocity with respect to the medium creates an energy "gradient." This is the true physical quantity behind the subtle temperature fluctuations in CMB data.

Logical conclusion: The CMB dipole is the cosmic "flow meter".

The observed amplitude of the CMB dipole is not our speed of movement, but rather an absolute indicator of **"how fast (0.955c) the spatial medium is flowing along the dipole axis."**

5. SOLVING THE HUBBLE TENSION

When considering the ultrafast flow of the medium, the redshift of distant quasars includes a "flow-away effect by the medium." Correcting for this naturally resolves the discrepancy between the Hubble constants of the near- and far-away universes.

6. FINAL UNIFICATION

Quasar phase lag, light stagnation, CMB dipoles—all these observational facts can be unified and explained by a single geometry: a helical auger flow on a trillion-light-year scale.

[Literal Japanese Translation]
● Redefining **CMB Dipole Anisotropy:** ● **The anisotropy is not due to our motion, but** to the 0.955c flow:
● **The opposite direction acts** as an energy vacuum where light stagnates : ● The CMB is a "speedometer" **of the space medium:**

DISCUSSION: BEYOND THE 0.046 CONVERGENCE — THE ABYSS OF THE 0.001C DIFFERENTIAL

6.1 THE PRECISION CRISIS AND THEORETICAL DEPTH

In this analysis, the media flow rate$v_m = 0.955$cand residual speed $V_{obs} = 0.046c$After examining the relationship, a simple offsetting calculation was found.1.000$c - 0.954c = 0.046c$**The difference of 0.001c** in this case was highlighted.

$$\delta = v_{medium}(0.955c) - v_{calc}(0.954c) = 0.001c$$

This subtle "play" is not a calculation error, but rather a matter of the spacetime medium18.2°These are physical traces of higher-order vibrational components that occur when flowing along a spiral slope.

6.2 THE SECOND-ORDER CORRECTION TERM

Correction items$\alpha = 0.046$While this governs the steady-state twisting of space,0.001The difference suggests an "elasticity" or "quantum slip" in the spacetime medium. This provides the basis for the uncertainty (jitter) that occurs when light does not completely stop at the edge of the medium but "penetrates" the event horizon of information.

> **[Reflections into the Abyss: 0.001 Residual Energy**
>]0.001cCould this difference be the true origin of the minute temperature fluctuations (anisotropy) in the cosmic microwave background radiation? Hidden behind the static 0.046 (geometry), this dynamic 0.001 (fluctuation) may be creating the gravitational "stagnation" that serves as the seed for galaxy formation.

6.3 IMPACT ON QUASAR OFS CONSISTENCY

The subtle long-term phase-locking deviations observed in high OFS targets such as 3C 273 are due to this0.001It is reinterpreted as a dynamic variation of the components. The "gap" between the certainty that the medium flow velocity is $0.955c$ and the 0.954 needed to computationally derive 0.046 is the point of contact that connects theory to real-world complex systems.

6.4 FINAL GEOMETRIC INTEGRATION

When the pitch angle of the spiral varies locally,0.001This functions as the "expansion and contraction (breathing)" of the spiral. This can be seen as evidence that our observable universe is currently "pulsating" within a spiral auger on a scale of one trillion light-years, accompanied by these minute differences in velocity.

> **[Literal Japanese Translation]**
> ● The abyss of **the 0.001c Differential:**
> ● **The 0.001 variance is a** physical trace of higher-order vibration:
> ● **Elasticity or quantum slip of the** spacetime medium :
> ● The "gap" is evidence **that** our universe is pulsating.

MATHEMATICAL FORMALIZATION: THE COSMIC PULSATION CONSTANT Z

7.1 BEYOND STATIC GEOMETRY

Correction items$\alpha = 0.046$This defines the static refractive index in a helical structure on a scale of one trillion light-years. However, the effective velocity of the medium$v_m = 0.955c$This can be calculated by working backward from the conditions for light stagnation.$0.954c$The subtle differences that arise between them suggest that spacetime is an "elastic body."

7.2 DEFINITION OF Z

We$0.001c$The difference is a constant that governs the dynamic pulsation of the universe.ζDefined as (Zeta).

$$\zeta = \frac{v_{eff} - (c - \alpha c)}{c} \approx 0.001$$

Here$v_{eff}$is the effective flow velocity,αThis is a geometric correction term.

[Physics of the Abyss:0.001The dynamic universe created by

static0.046This determines the angle of the spiral "rail," and the dynamic$\zeta = 0.001$This constant determines the "vibration (beat)" of the medium running on that rail. This constant physically confirms that the universe is not a perfect vacuum but has a subtle "viscosity" or "quantum slip." This is the true nature of the fluctuating components of "accelerated expansion of the universe" and "dark energy" in modern astronomy.

7.3 PULSATION AND GALAXY CLUSTER FORMATION

This pulsation constantζThis manifests as "standing wave fluctuations" at the boundaries of giant voids and filament structures.0.001This velocity difference is a sufficient threshold to create gravitational stagnation and becomes a dynamic seed that determines the density distribution of galaxy clusters.

7.4 CONCLUSION OF DISCUSSION

$\alpha = 0.046$If that is the "form" of the universe,$\zeta = 0.001$This is its "pulse." The integration of these two constants elevates the spacetime refraction model (WRM) into a complete dynamic theory that accurately describes the anomalies of the entire observed universe.

[Literal Japanese Translation]

- **The Cosmic Pulsation Constantζ:** Cosmic pulsation constantζ。
- **If α is the "form" of the universe, ζ is its "heartbeat":** αIf that is the "appearance" of the universe,ζThis is its "heartbeat."
- **This constant accounts for the cosmic** "viscosity" or "quantum slip. "
- The final integration of static geometry and dynamic flow **.**

TECHNICAL NOTE: THE ONTOLOGICAL NECESSITY OF VISCOSITY IN STANDING WAVES

1. VISCOSITY AS COHESIVE FORCE

For standing waves to exist, each point in the medium must have a "binding force" that allows it to exchange energy with neighboring points. In a perfectly inviscid fluid (superfluid), energy passes through without dissipation, but there is no constraint force to maintain a specific "shape (helical structure)".

2. RESISTANCE VS. MAINTENANCE

Here, "viscosity" is synonymous with both "resistance" that hinders movement and "tension" that binds the medium together in a spiral. Viscous term in the Navier-Stokes equations$\eta \nabla^2 v$In WRM, this functions as a "geometric stiffness" that fixes the helical pitch.

[Physical interpretation: Why viscosity is necessary]
A "standing wave" is a state of equilibrium between propagating waves and reflected waves. A spiral auger on a scale of one trillion light-years can maintain its shape because the medium has "viscosity (internal friction)," which "retains" energy in space and structures it. If it were non-viscous, the spiral would instantly unravel and collapse into straight-line radiation.

3. THE 0.001 DIFFERENTIAL (Z)

Cosmic pulsation constant$\zeta = 0.001$This represents a numerical representation of the "transmission delay" caused by viscosity. This minute deviation from an ideal standing wave is the driving force (torque) that keeps the universe from being completely still but from constantly pulsating dynamically.

4. CONCLUSION

Therefore, viscosity is not an enemy that attenuates standing waves, but rather a prerequisite for standing waves to "exist" in this universe. While synonymous with resistance, it plays the role of a "framework" that supports the structure.

[Literal Japanese Translation]
- **Viscosity as Cohesive Force:** Viscosity as a cohesive force.
- **Viscosity is a prerequisite for the existence of standing waves:** Viscosity is a prerequisite for the existence of standing waves.
- **Resistance is synonymous with the "skeletal frame" of the structure:** Resistance is synonymous with the "skeletal frame" of the structure.
- **The 0.001 variance represents the "torque" of cosmic pulsation:** The 0.001 variance represents the "torque" of cosmic pulsation.

APPENDIX: THE ENTROPY PARADOX IN A CYCLIC HELICAL UNIVERSE

A.1 THE THERMAL ACCUMULATION PROBLEM

Standing waves$0.001c$If a substance has "viscosity," a portion of its energy is inevitably converted into heat. In a closed circulating system (10,000 connected units), if this thermal entropy continues to accumulate, the universe will eventually die from heat.

A.2 MULTI-UNIVERSE COOLING MECHANISM

From a multi-universe perspective, it is hypothesized that a "micro-cold void (outside of information)" exists outside our spiral, and that a cooling mechanism exists to expel entropy into it. This could manifest as a phase transition phenomenon through the nodes (constrictions) of the spiral.

[Three Scenarios for Entropy Elimination]
1. **Phase Transition Reset:** At the end of the spiral (10,000th), the heated medium undergoes a "phase transition" back to quantum zero-point oscillations, returning to the starting point.
2. **Neutralization by External Waves:** Waves with opposite phases from other universes (multiverses) cancel out the thermal noise in our universe, reconstructing standing waves.
3. **Evaporation of Information:** $0.046c$The "leaking" components are carrying away the accumulated entropy to the outside of the outer shell of the universe.

A.3 THE ROLE OF THE 0.001 GAP (Ζ)

Cosmic pulsation constant$\zeta = 0.001$This may represent the thermal fluctuations themselves. Without this "play," the universe would be unable to release entropy and would instantly become thermally saturated.

A.4 UNANSWERED QUESTIONS

Is this universe a one-way ticket that only gets hotter, or is it a sophisticated self-cooling engine? The answer will remain shrouded in mystery until we can observe the physical phenomena at the boundaries (nodes) of the spiral units.

[Literal Japanese Translation]
- **The Entropy Paradox:** The paradox of entropy.
- **Heat accumulation would lead to the Heat Death:** Heat accumulation would lead to death by heat.
- **The 0.001 gap allows for thermal fluctuation:** The 0.001 gap allows for thermal fluctuation.
- **Is the universe a one-way ticket or a self-cooling engine?:** Is the universe a one-way ticket or a self-cooling engine?